Given:

To perform Exploratory Data Analysis (EDA) on the Titanic dataset using Python to understand data distribution, identify patterns, detect outliers, and analyze relationships between features through statistical summaries and visualizations.

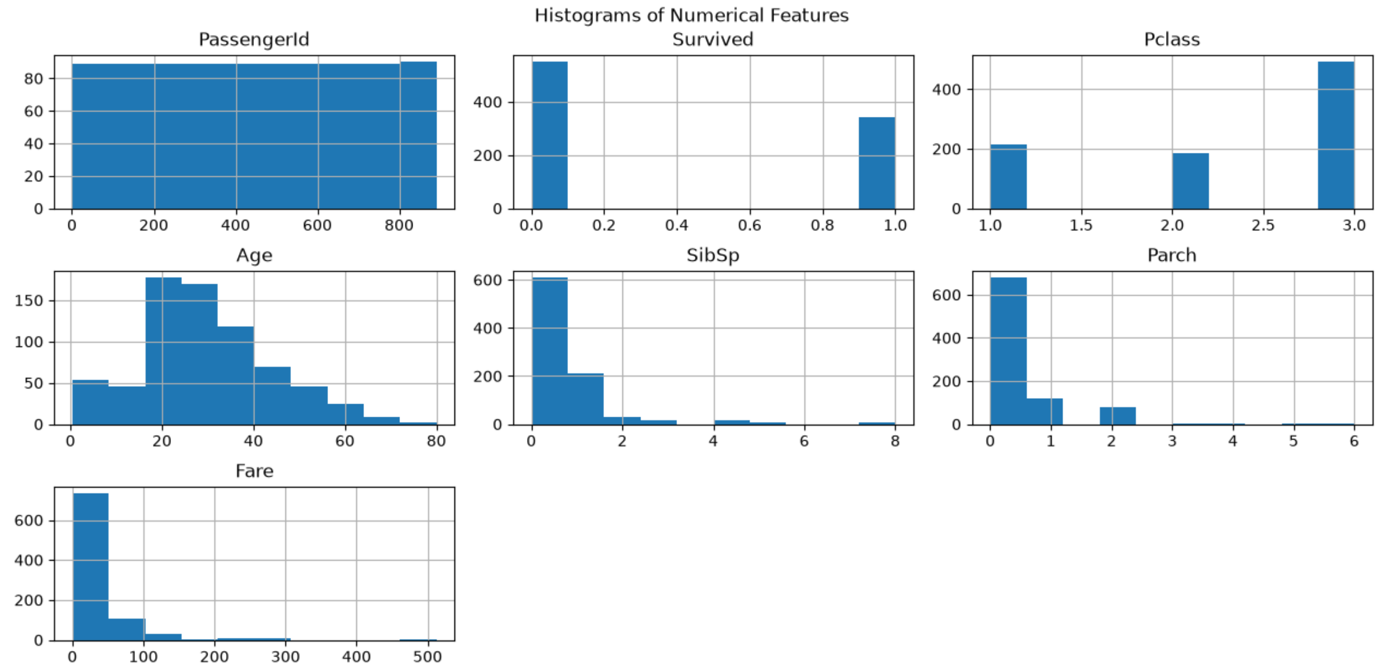
DATASET: Click to Download Titanic-Dataset

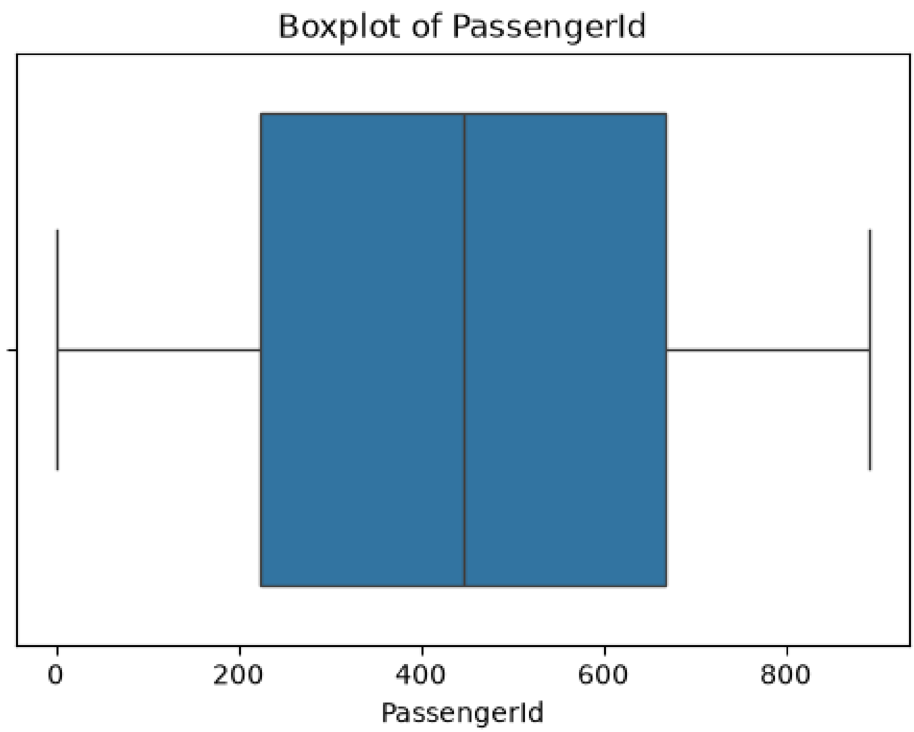
Data_Cleanin.py

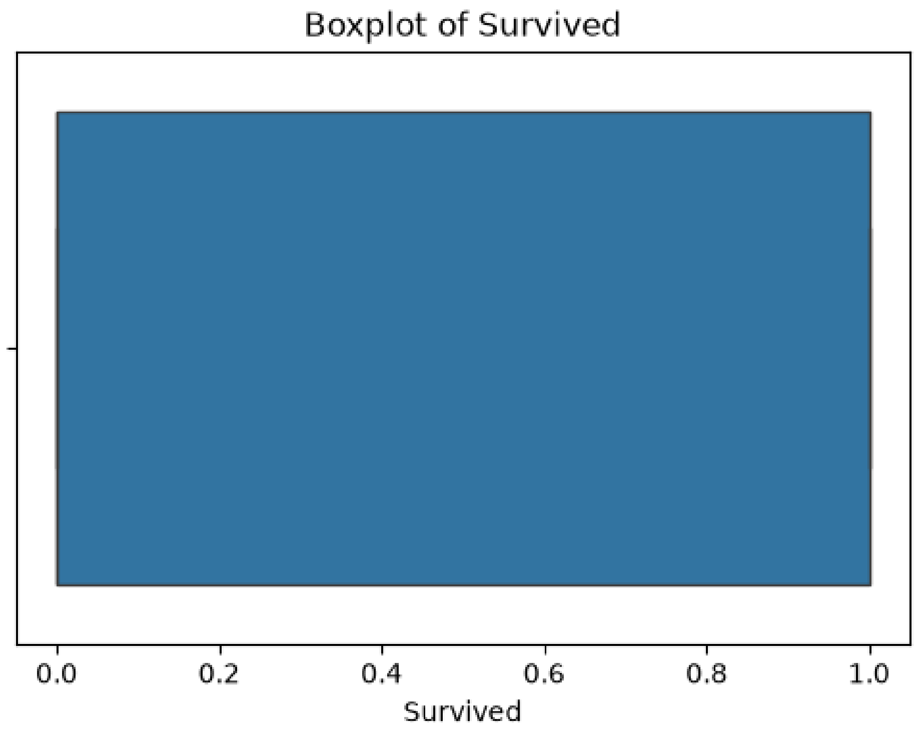
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
df = pd.read_csv("Titanic-Dataset.csv")
print("First Five Rows")
print(df.head())
print("\nDataset Information")
print(df.info())
print("\nSummary Statistics")
print(df.describe(include='all'))
print("\nMissing Values")
print(df.isnull().sum())
numeric_columns = df.select_dtypes(include=['int64', 'float64']).columns
df[numeric_columns].hist(figsize=(12,10))
plt.suptitle("Histograms of Numerical Features")
plt.tight_layout()
plt.show()
for col in numeric_columns:
    plt.figure(figsize=(6,4))
    sns.boxplot(x=df[col])
    plt.title(f"Boxplot of {col}")
    plt.show()
numeric_df = df.select_dtypes(include=['number'])
plt.figure(figsize=(10,8))
sns.heatmap(numeric_df.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.show()
sns.pairplot(df[['Age','Fare','Pclass','Survived']].dropna(),hue='Survived')
plt.show()
plt.figure(figsize=(6,4))
sns.countplot(x='Survived', data=df)
plt.title("Passenger Survival Count")
plt.show()
plt.figure(figsize=(6,4))
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title("Survival by Gender")
plt.show()
plt.figure(figsize=(6,4))
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title("Survival by Passenger Class")
plt.show()
fig = px.histogram(df,x="Age",color="Survived",title="Age Distribution by Survival")
fig.show()
```

Output:

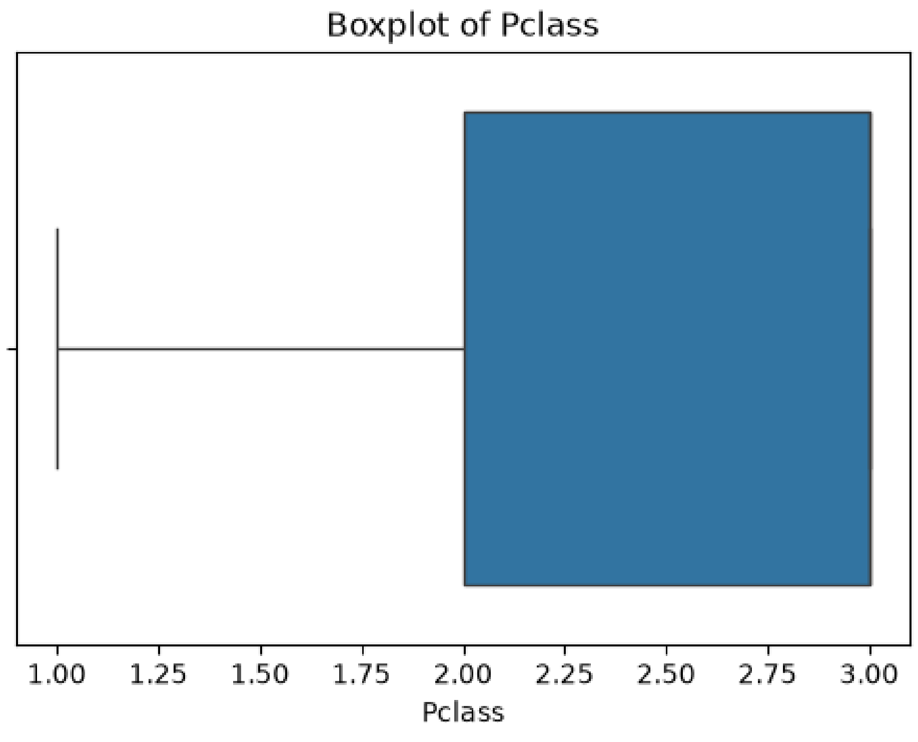
```
First Five Rows
   PassengerId  Survived  Pclass                                               Name     Sex   Age  SibSp  Parch            Ticket     Fare Cabin Embarked
0            1         0       3                            Braund, Mr. Owen Harris    male  22.0      1      0         A/5 21171   7.2500   NaN        S
1            2         1       1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1      0          PC 17599  71.2833   C85        C
2            3         1       3                             Heikkinen, Miss. Laina  female  26.0      0      0  STON/O2. 3101282   7.9250   NaN        S
3            4         1       1       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1      0            113803  53.1000  C123        S
4            5         0       3                           Allen, Mr. William Henry    male  35.0      0      0            373450   8.0500   NaN        S
Dataset Information
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    str
 4   Sex          891 non-null    str
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    str
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str
 11  Embarked     889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
None
Summary Statistics
        PassengerId    Survived      Pclass                     Name   Sex         Age       SibSp       Parch  Ticket        Fare Cabin Embarked
count    891.000000  891.000000  891.000000                      891   891  714.000000  891.000000  891.000000     891  891.000000   204      889
unique          NaN         NaN         NaN                      891     2         NaN         NaN         NaN     681         NaN   147        3
top             NaN         NaN         NaN  Braund, Mr. Owen Harris  male         NaN         NaN         NaN  347082         NaN    G6        S
freq            NaN         NaN         NaN                        1   577         NaN         NaN         NaN       7         NaN     4      644
mean     446.000000    0.383838    2.308642                      NaN   NaN   29.699118    0.523008    0.381594     NaN   32.204208   NaN      NaN
std      257.353842    0.486592    0.836071                      NaN   NaN   14.526497    1.102743    0.806057     NaN   49.693429   NaN      NaN
min        1.000000    0.000000    1.000000                      NaN   NaN    0.420000    0.000000    0.000000     NaN    0.000000   NaN      NaN
25%      223.500000    0.000000    2.000000                      NaN   NaN   20.125000    0.000000    0.000000     NaN    7.910400   NaN      NaN
50%      446.000000    0.000000    3.000000                      NaN   NaN   28.000000    0.000000    0.000000     NaN   14.454200   NaN      NaN
75%      668.500000    1.000000    3.000000                      NaN   NaN   38.000000    1.000000    0.000000     NaN   31.000000   NaN      NaN
max      891.000000    1.000000    3.000000                      NaN   NaN   80.000000    8.000000    6.000000     NaN  512.329200   NaN      NaN
Missing Values
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```


Histograms of Numerical Features
PassengerId
Survived
Pclass
Age
SibSp
Parch
Fare

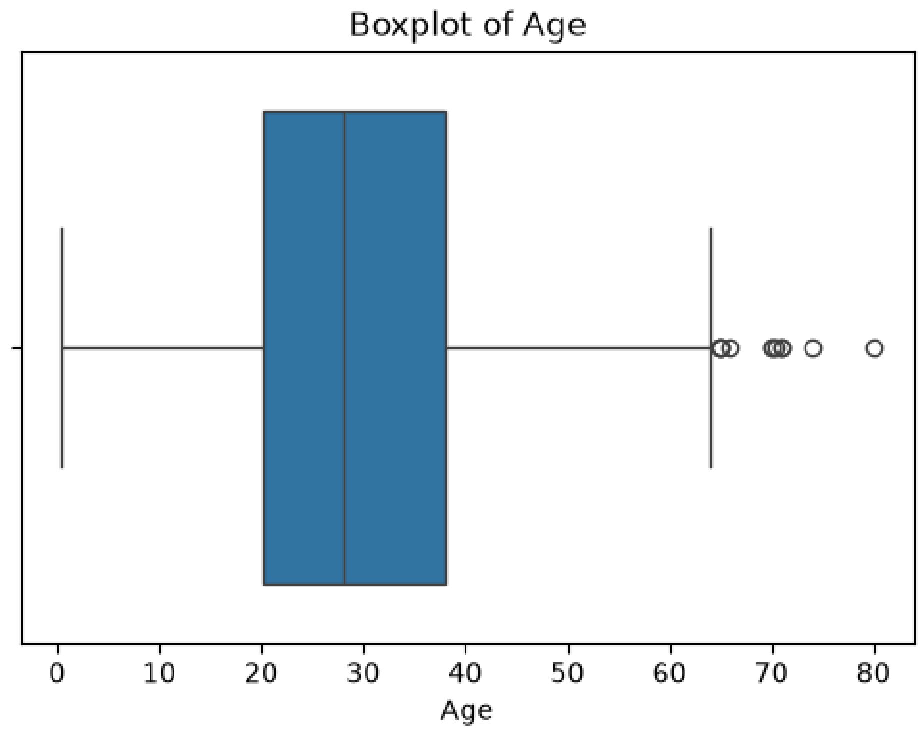

Boxplot of PassengerId
PassengerId

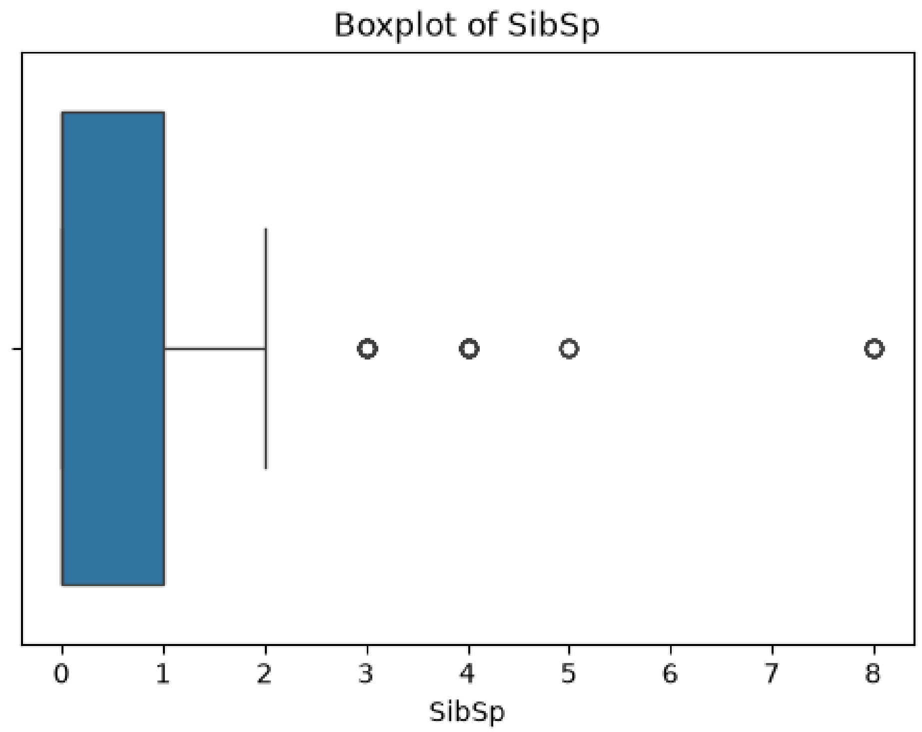


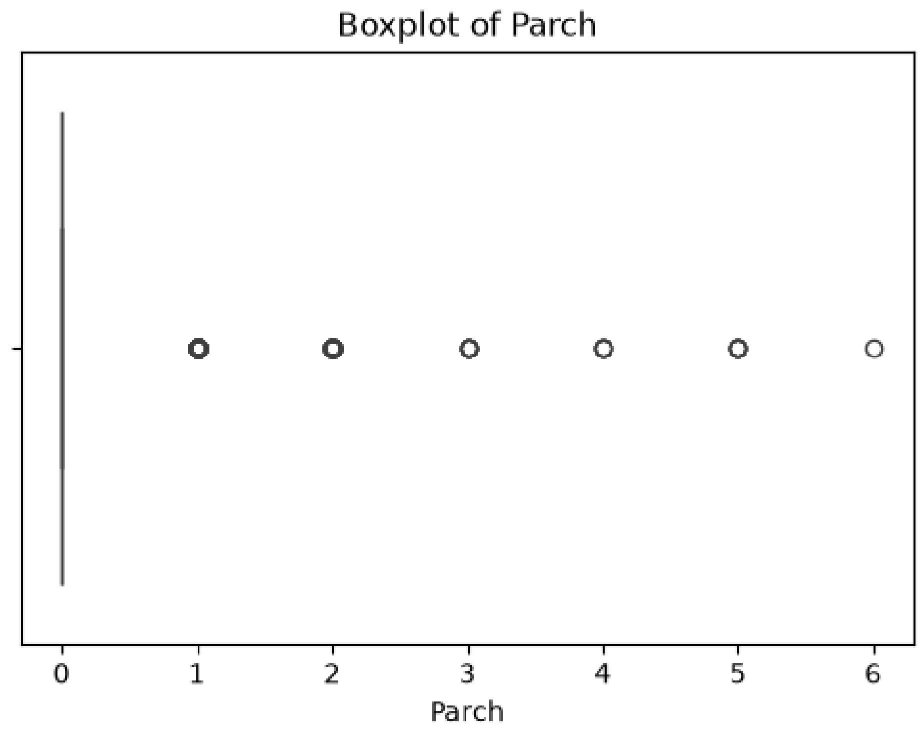
Boxplot of Survived
0.0
0.2
0.4
0.6
0.8
1.0
Survived

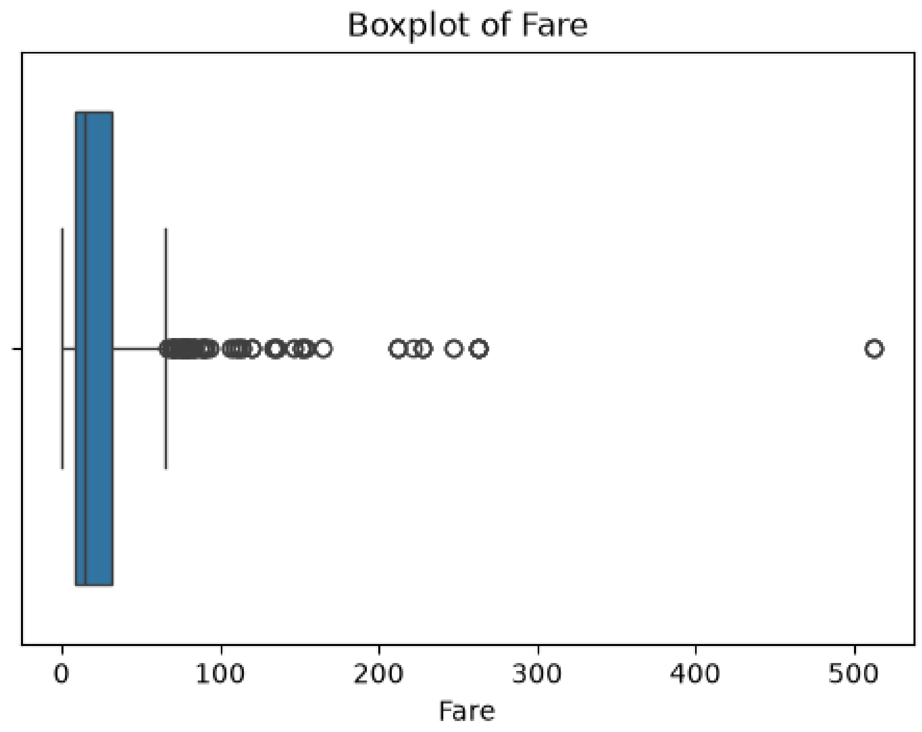


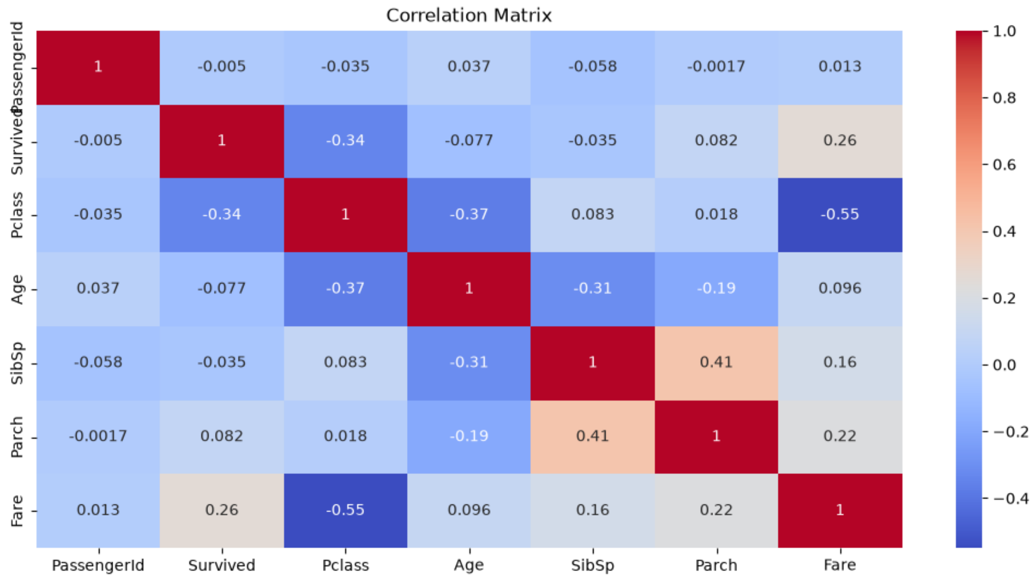
Boxplot of Pclass
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00
Pclass

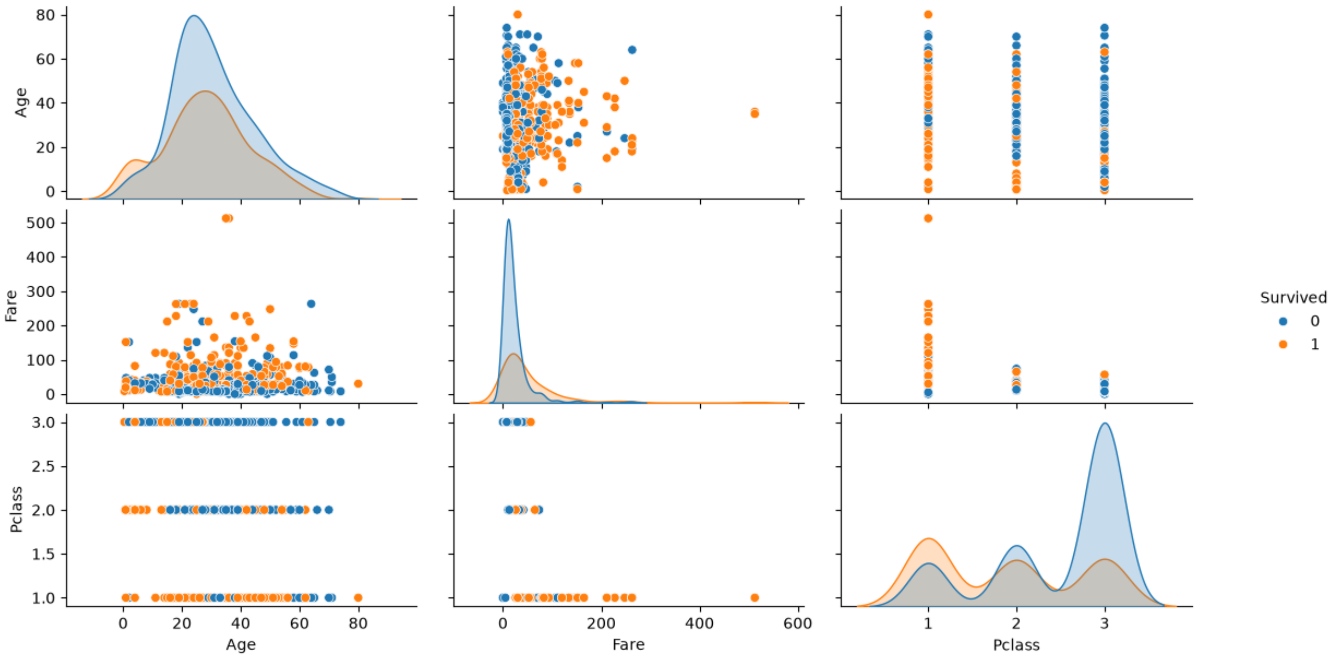

Boxplot of Age
0
10
20
30
40
50
60
70
80
Age

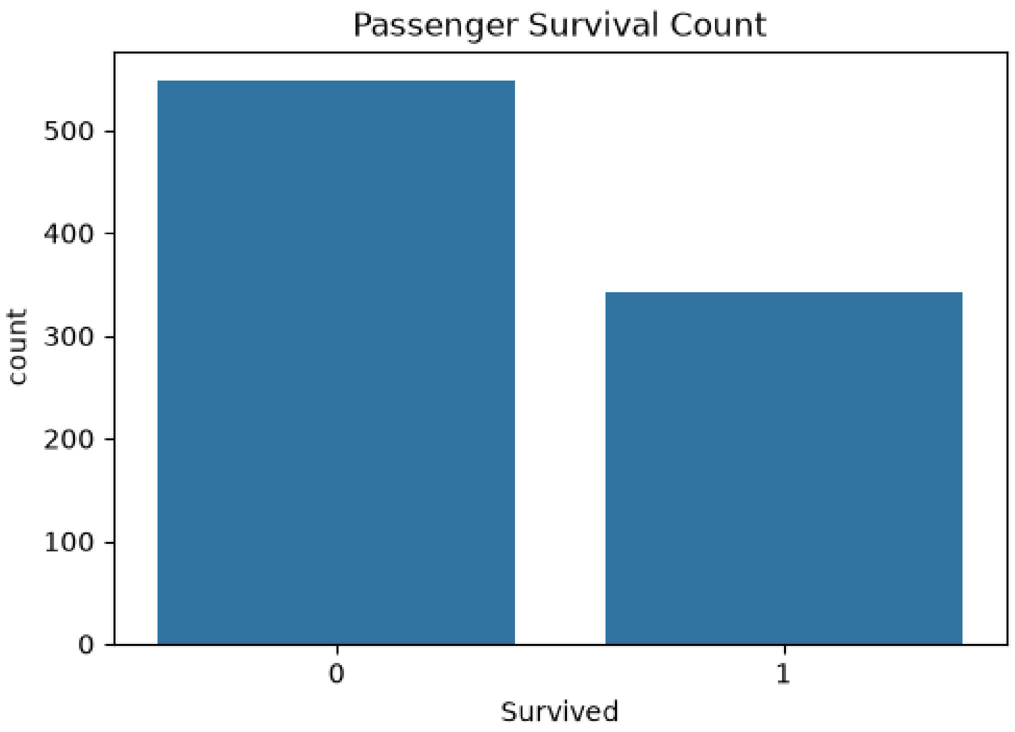

Boxplot of SibSp
0
1
2
3
4
5
6
7
8
SibSp

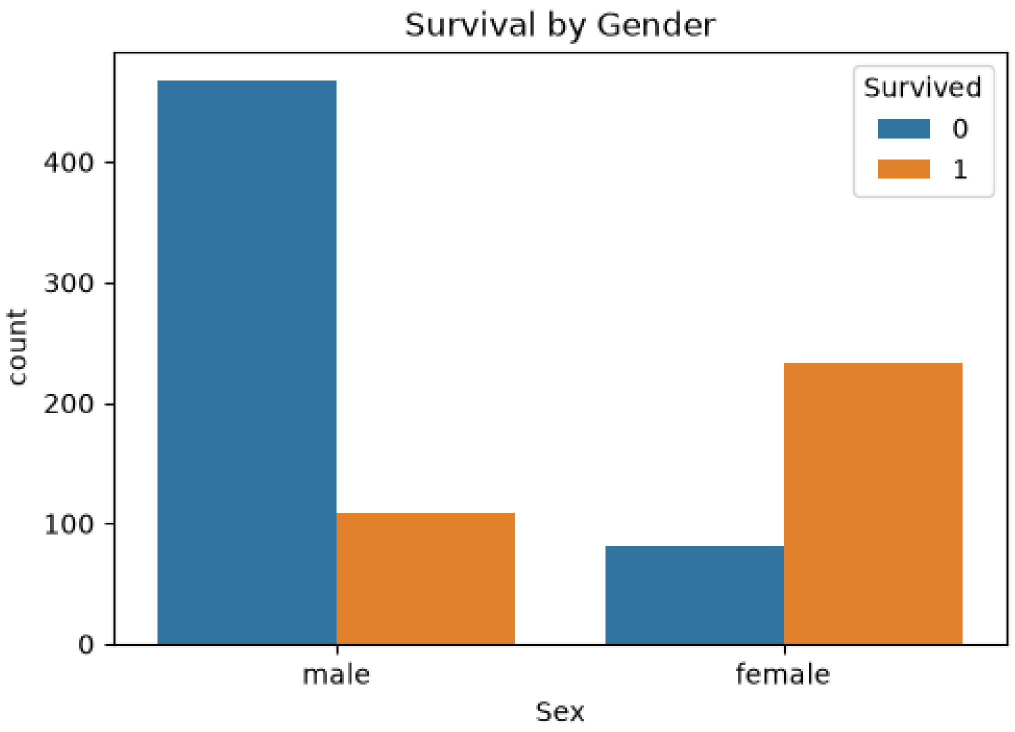

Boxplot of Parch
0
1
2
3
4
5
6
Parch

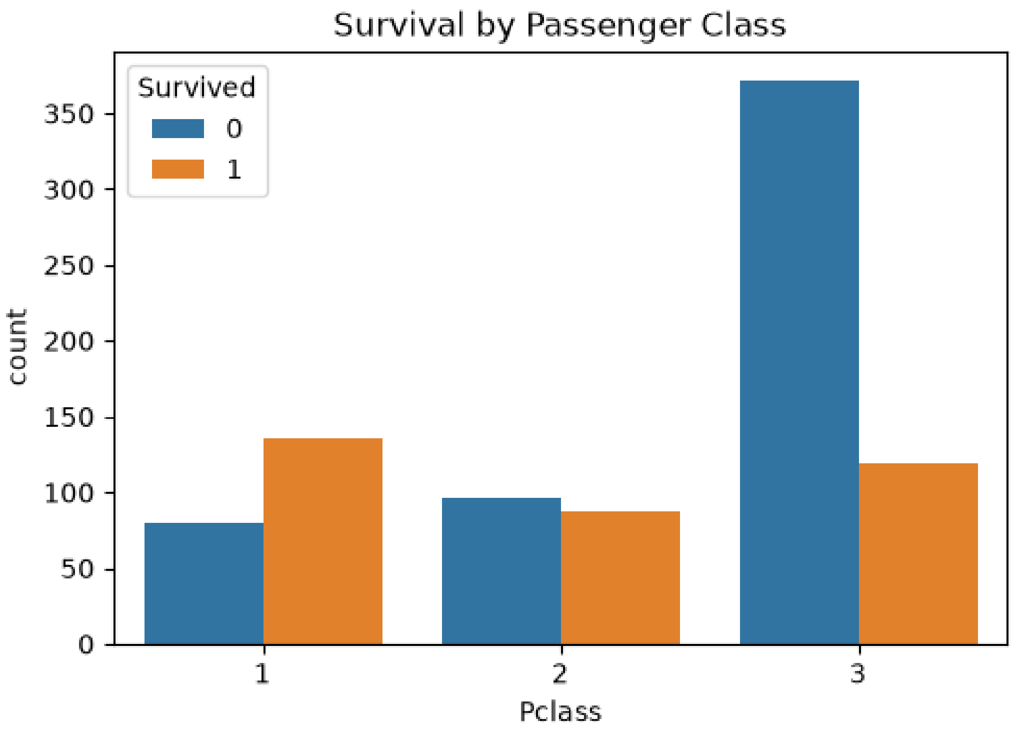

Boxplot of Fare
0
100
200
300
400
500
Fare


Correlation Matrix
PassengerId
Survived
Pclass
Age
SibSp
Parch
Fare
1 -0.005 -0.035 0.037 -0.058 -0.0017 0.013
-0.005 1 -0.34 -0.077 -0.035 0.082 0.26
-0.035 -0.34 1 -0.37 0.083 0.018 -0.55
0.037 -0.077 -0.37 1 -0.31 -0.19 0.096
-0.058 -0.035 0.083 -0.31 1 0.41 0.16
-0.0017 0.082 0.018 -0.19 0.41 1 0.22
0.013 0.26 -0.55 0.096 0.16 0.22 1
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4


Age
Fare
Pclass
Survived
0
1


Passenger Survival Count
count
0
100
200
300
400
500
0
1
Survived


Survival by Gender
count
0
100
200
300
400
Survived
0
1
male
female
Sex



Age Distribution by Survival